Welcome to

Become a Python Developer

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List Problems

Find the Sum of All Elements in a List

```
numbers = [1, 2, 3, 4, 5]
total = 0

for number in numbers:
        total += number
print(total)
```

Remove Duplicates

```
my_list = [1, 2, 2, 3, 4, 4, 5]
unique_list = []

for item in my_list:

        if item not in unique_list:
        unique_list.append(item)

print(unique_list)
```

Find the Maximum

```
numbers = [7, 2, 9, 4, 1, 5]
max_element = numbers[0]

for number in numbers:
        if number > max_element:
max_element =number

print(max_element)
```

Tuple Problems

Tuple addition

```
t1 = (1, 2, 3)
t2 = (4, 5, 6)
summed_tuple = tuple(t1[i] + t2[i]
for i in range(len(t1)))
print(summed_tuple)
```

Frequency count

```
t = (1, 2, 3, 1, 2, 1, 4)
frequency_dict = {}
for element in t:
        if element in frequency_dict:
        frequency_dict[element] += 1
        else:
        frequency_dict[element] = 1
print(frequency_dict)
```

Merge two tuples

```
tuple_a = (1, 2, 3)
tuple_b = (4, 5, 6)

merged_tuple = tuple_a + tuple_b

print(merged_tuple)
```

Dictionary Problems

Merge Two dictionaries

```
dict_a = {'a': 1, 'b': 2}
dict_b = {'c': 3, 'd': 4}
 merged_dict = {**dict_a, **dict_b}
print(merged_dict)
```

Keys with maximum value

```
scores = {'Alice': 90, 'Bob': 85, 'Charlie': 90, 'David': 88}
max_value = max(scores.values())
max_keys = [key for key, value in scores.items() if value == max_value]
print(max_keys)
```

Remove a key

```
my_dict = {'a': 1, 'b': 2, 'c': 3}
key_to_remove = 'b‘
if key_to_remove in my_dict:
        del my_dict[key_to_remove]
    print(my_dict)
```

Strings Problems

Reverse a string

```
text = "Hello, World!“
reversed_str = text[::-1]
print(reversed_str)
```

Check for palindrome

```
text = "racecar“
text = text.lower()  # Convert to lowercase for case insensitivity
reversed_str = text[::-1]
is_palindrome = text == reversed_str
print(is_palindrome)
```

Count the occurrences of each character

```
text = "programming“
char_count = {}
for char in text:
        if char in char_count:
char_count[char] += 1
        else:
                char_count[char] = 1
print(char_count)
```